

Report: Big Data Analytics Project Workflow

1. Project Overview

This report outlines the structure of the **WeeklyPatentAnalytics** project hosted on GitHub. The project is a series of big data analytics tasks implemented entirely in **Jupyter Notebook** using **Apache Spark (PySpark)**. The repository contains a structured sequence of notebooks reflecting a workflow designed to process, analyze, and summarize **weekly patent application data**.

Repository Details:

- **Name:** WeeklyPatentAnalytics
 - **Language:** Jupyter Notebook (100.0%)
 - **Description:** Big data analysis of weekly patent filings using PySpark.
 - **Activity:** The repository currently has 0 stars, 0 watchers, and 0 forks.
 - **Releases/Packages:** No official releases or packages have been published.
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2. Inferred Analysis Workflow

The project is structured into a series of numbered notebooks, indicating a step-by-step approach to working with large-scale patent datasets. It demonstrates how to ingest, clean, analyze, and report on data using PySpark's powerful capabilities.

1. Data Transformation Using PySpark RDD

The process begins with loading and transforming patent data using PySpark RDDs. This allows for scalable processing of large CSV or JSON files typically provided by patent offices such as USPTO or WIPO.

2. Collect(), Sample(), and TakeSample() Operations

These operations are used to retrieve small portions of the data for exploration and validation. Sampling helps in testing logic before applying transformations to the entire dataset.

3. Exploring Structure and Contents of a Patent Data File

This step focuses on schema discovery and exploratory data analysis (EDA). The data, often in CSV or JSON format, is inspected for key fields such as:

- Filing Date
 - Technology Classification (IPC/CPC)
 - Applicant Country
 - Assignee Name
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4. Viewing Data and Selecting Columns

Relevant fields are selected from the dataset to narrow down the analysis. This includes filtering by filing week, technology fields, or inventor location to drive insights from specific perspectives.

5. Analytical Operations on Weekly Patent Data

This stage performs operations like:

- Weekly counts of new patent applications
- Top technology categories filed
- Leading countries and applicants
- Trends in innovation by sector

Aggregations, filtering, and sorting are used to draw meaningful conclusions.

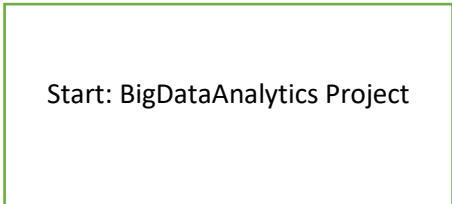
6. Weekly Patent Filing Insights (Mini Project)

The project culminates in a mini-report or dashboard that:

- Summarizes the number of applications filed in the week
- Highlights top 5 technology fields and countries
- Identifies top assignees or organizations
- Offers trend insights over time if historical data is available

3. Workflow Flowchart

The following flowchart visualizes the sequential process inferred from the repository's file structure.



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Start: BigDataAnalytics Project

1. DataTransformation Using
PySpark RDD

2. Collect() Operation with RDD
Operation

3. Sample() and TakeSample()
Methods

4. Exploring Structure And
Contents of CSV File

5. Viewing Data And Selecting
Columns Of CSV File

6. Analytical Operations on CSV
File

7. Weekly patent Analysis(Mini
Project)

End of Analysis